# LEMMA COVERINGS SITE

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*Editorial note.* All mathematical statement and proof text is reproduced verbatim from the author's chapter sets.tex of the Stacks Project, fixed at commit a04446e57ec1fbc252a871afcec7752fb2807b14. The pinned preamble is supplied as its own file; the document wrapper class, the xr-hyper wiring and the \externaldocument list are omitted because the assembler supplies the class.

*Editorial note.* Numbering inside this standalone excerpt is editorial, not the full Stacks book print numbering.

*Editorial note.* The candidate was previously judged H2 by the initial-corpus stage; that judgement and its reason are reused verbatim and were not re-derived here. Original candidate key: S001-lemma-coverings-site.

*Editorial note.* Reference labels in the proof resolve to navigation pointers to the original author source; these pointers do not reproduce proofs of the prerequisites.

*Allowed standard prerequisites (editorial).*

- Prerequisites named by the author inside the counted span. The counted span is reproduced verbatim, including the author's own citations. Its prerequisites are the earlier results of the same fixed Stacks Project version that the author cites there and in the chapter's lead-in.

*Source reference.* Original definition 00VH: sources/sites.tex, source lines 679–679. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

*Source reference.* Original proposition 05N3: sources/sets.tex, source lines 217–217. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

**Lemma 0.1.** *With notations as above. Let $Cov_0 \subset Cov(\mathcal{C})$ be a set contained in $Cov(\mathcal{C})$. There exist a cardinal $\kappa$ and a limit ordinal $\alpha$ with the following properties:*

1. *We have $Cov_0 \subset Cov(\mathcal{C})_{\kappa,\alpha}$.*
2. *The set of coverings $Cov(\mathcal{C})_{\kappa,\alpha}$ satisfies (1), (2), and (3) of Sites, Definition 00VH (see above). In other words $(\mathcal{C}, Cov(\mathcal{C})_{\kappa,\alpha})$ is a site.*
3. *Every covering in $Cov(\mathcal{C})$ is combinatorially equivalent to a covering in $Cov(\mathcal{C})_{\kappa,\alpha}$.*

*Proof.* To prove this, we first consider the set $\mathcal{S}$ of all sets of morphisms of $\mathcal{C}$ with fixed target. In other words, an element of $\mathcal{S}$ is a subset $T$ of $\mathrm{Arrows}(\mathcal{C})$ such that all elements of $T$ have the same target. Given a family $\mathcal{U} = \{\varphi_i : U_i \to U\}_{i \in I}$ of morphisms with fixed target, we define $Supp(\mathcal{U}) = \{\varphi \in \mathrm{Arrows}(\mathcal{C}) \mid \exists i \in I, \varphi =$

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$\varphi_i\}$. Note that two families $\mathcal{U} = \{\varphi_i : U_i \to U\}_{i\in I}$ and $\mathcal{V} = \{V_j \to V\}_{j\in J}$ are combinatorially equivalent if and only if $Supp(\mathcal{U}) = Supp(\mathcal{V})$. Next, we define $\mathcal{S}_\tau \subset \mathcal{S}$ to be the subset $\mathcal{S}_\tau = \{T \in \mathcal{S} \mid \exists\ \mathcal{U} \in \mathrm{Cov}(\mathcal{C})\ T = Supp(\mathcal{U})\}$. For every element $T \in \mathcal{S}_\tau$, set $\beta(T)$ to equal the least ordinal $\beta$ such that there exists a $\mathcal{U} \in \mathrm{Cov}(\mathcal{C})_\beta$ such that $T = \mathrm{Supp}(\mathcal{U})$. Finally, set $\beta_0 = \sup_{T\in S_\tau} \beta(T)$. At this point it follows that every $\mathcal{U} \in \mathrm{Cov}(\mathcal{C})$ is combinatorially equivalent to some element of $\mathrm{Cov}(\mathcal{C})_{\beta_0}$.

Let $\kappa$ be the maximum of $\aleph_0$, the cardinality $|\mathrm{Arrows}(\mathcal{C})|$,

$$\sup_{\{U_i\to U\}_{i\in I}\in \mathrm{Cov}(\mathcal{C})_{\beta_0}} |I|, \quad \text{and} \quad \sup_{\{U_i\to U\}_{i\in I}\in \mathrm{Cov}_0} |I|.$$

Since $\kappa$ is an infinite cardinal, we have $\kappa \otimes \kappa = \kappa$. Note that obviously $\mathrm{Cov}(\mathcal{C})_{\beta_0} = \mathrm{Cov}(\mathcal{C})_{\kappa,\beta_0}$.

We define, by transfinite induction, a function $f$ which associates to every ordinal an ordinal as follows. Let $f(0) = 0$. Given $f(\alpha)$, we define $f(\alpha+1)$ to be the least ordinal $\beta$ such that the following hold:

(1) We have $\alpha + 1 \leq \beta$ and $f(\alpha) \leq \beta$.
(2) If $\{U_i \to U\}_{i\in I} \in \mathrm{Cov}(\mathcal{C})_{\kappa,f(\alpha)}$ and for each $i$ we have $\{W_{ij} \to U_i\}_{j\in J_i} \in \mathrm{Cov}(\mathcal{C})_{\kappa,f(\alpha)}$, then $\{W_{ij} \to U\}_{i\in I, j\in J_i} \in \mathrm{Cov}(\mathcal{C})_{\kappa,\beta}$.
(3) If $\{U_i \to U\}_{i\in I} \in \mathrm{Cov}(\mathcal{C})_{\kappa,\alpha}$ and $W \to U$ is a morphism of $\mathcal{C}$, then $\{U_i \times_U W \to W\}_{i\in I} \in \mathrm{Cov}(\mathcal{C})_{\kappa,\beta}$.

To see $\beta$ exists we note that clearly the collection of all coverings $\{W_{ij} \to U\}$ and $\{U_i \times_U W \to W\}$ that occur in (2) and (3) form a set. Hence there is some ordinal $\beta$ such that $V_\beta$ contains all of these coverings. Moreover, the index set of the covering $\{W_{ij} \to U\}$ has cardinality $\sum_{i\in I} |J_i| \leq \kappa \otimes \kappa = \kappa$, and hence these coverings are contained in $\mathrm{Cov}(\mathcal{C})_{\kappa,\beta}$. Since every nonempty collection of ordinals has a least element we see that $f(\alpha+1)$ is well defined. Finally, if $\alpha$ is a limit ordinal, then we set $f(\alpha) = \sup_{\alpha'<\alpha} f(\alpha')$.

Pick an ordinal $\beta_1$ such that $\mathrm{Arrows}(\mathcal{C}) \subset V_{\beta_1}$, $\mathrm{Cov}_0 \subset V_{\beta_0}$, and $\beta_1 \geq \beta_0$. By construction $f(\beta_1) \geq \beta_1$ and we see that the same properties hold for $V_{f(\beta_1)}$. Moreover, as $f$ is nondecreasing this remains true for any $\beta \geq \beta_1$. Next, choose any ordinal $\beta_2 > \beta_1$ with cofinality $\mathrm{cf}(\beta_2) > \kappa$. This is possible since the cofinality of ordinals gets arbitrarily large, see Proposition 05N3. We claim that the pair $\kappa$, $\alpha = f(\beta_2)$ is a solution to the problem posed in the lemma.

The first and third property of the lemma holds by our choices of $\kappa$, $\beta_2 > \beta_1 > \beta_0$ above.

Since $\beta_2$ is a limit ordinal (as its cofinality is infinite) we get $f(\beta_2) = \sup_{\beta<\beta_2} f(\beta)$. Hence $\{f(\beta) \mid \beta < \beta_2\} \subset f(\beta_2)$ is a cofinal subset. Hence we see that

$$V_\alpha = V_{f(\beta_2)} = \bigcup\nolimits_{\beta<\beta_2} V_{f(\beta)}.$$

Now, let $\mathcal{U} \in \mathrm{Cov}_{\kappa,\alpha}$. We define $\beta(\mathcal{U})$ to be the least ordinal $\beta$ such that $\mathcal{U} \in \mathrm{Cov}_{\kappa,f(\beta)}$. By the above we see that always $\beta(\mathcal{U}) < \beta_2$.

We have to show properties (1), (2), and (3) defining a site hold for the pair $(\mathcal{C}, \mathrm{Cov}_{\kappa,\alpha})$. The first holds because by our choice of $\beta_2$ all arrows of $\mathcal{C}$ are contained in $V_{f(\beta_2)}$. For the third, we use that given a covering $\mathcal{U} = \{U_i \to U\}_{i\in I} \in \mathrm{Cov}(\mathcal{C})_{\kappa,\alpha}$ we have $\beta(\mathcal{U}) < \beta_2$ and hence any base change of $\mathcal{U}$ is by construction of $f$ contained in $\mathrm{Cov}(\mathcal{C})_{\kappa,f(\beta+1)}$ and hence in $\mathrm{Cov}(\mathcal{C})_{\kappa,\alpha}$.

Finally, for the second condition, suppose that $\{U_i \to U\}_{i\in I} \in \mathrm{Cov}(\mathcal{C})_{\kappa,f(\alpha)}$ and for each $i$ we have $\mathcal{W}_i = \{W_{ij} \to U_i\}_{j\in J_i} \in \mathrm{Cov}(\mathcal{C})_{\kappa,f(\alpha)}$. Consider the function $I \to \beta_2$, $i \mapsto \beta(\mathcal{W}_i)$. Since the cofinality of $\beta_2$ is $> \kappa \geq |I|$ the image of this function

cannot be a cofinal subset. Hence there exists a $\beta < \beta_1$ such that $\mathcal{W}_i \in \mathrm{Cov}_{\kappa,f(\beta)}$ for all $i \in I$. It follows that the covering $\{W_{ij} \to U\}_{i\in I, j\in J_i}$ is an element of $\mathrm{Cov}(\mathcal{C})_{\kappa,f(\beta+1)} \subset \mathrm{Cov}(\mathcal{C})_{\kappa,\alpha}$ as desired. $\square$